



3.8.4 Wrap-Up: Reflection

1. Circle the icon that best describes your understanding of each concept.



I need help.



I got it.



I could help someone.

Using <i>substitution</i> as a reason in a proof.	  
Using the <i>transitive property of congruence</i> as a reason in a proof.	  
Using the <i>definition of congruence</i> as a reason in a proof.	  
Understanding what angles are congruent when two lines are parallel.	  
Understanding what angles are supplementary when two lines are parallel.	  



Unit 3, Lesson 9: Angle Relationship Proofs – Part 2



Benchmarks

MA.912.GR.1.1 Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of lines and angles.

MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.



Learning Target

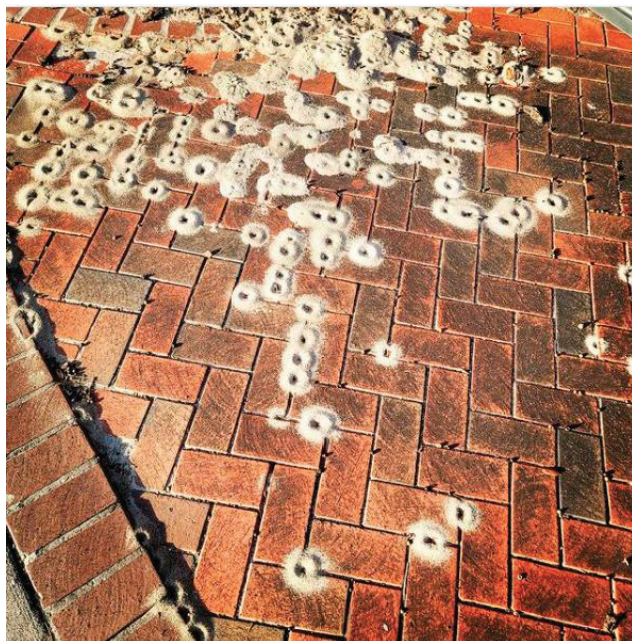
- ☐ I can prove relationships using parallel lines and their transversals.



3.9.1 Warm-Up: Notice and Wonder

A photo of ant hills taken in Gainesville, Florida is shown.

1. What do you notice?



2. What do you wonder?

3. Write a caption for the photo.



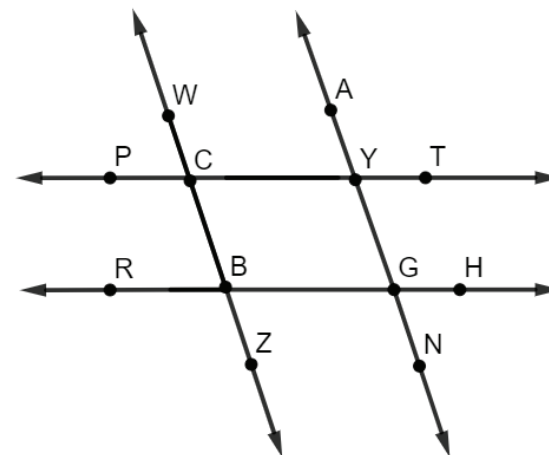
3.9.2 Activity: Card Sort

1. With your group, sort through the statements and reasons for each proof. Then, write each proof in the tables provided.

Proof A

Given: $\overleftrightarrow{WZ} \parallel \overleftrightarrow{AN}$,
 $\angle WCY \cong \angle YGH$

Prove: $\overleftrightarrow{PT} \parallel \overleftrightarrow{RH}$

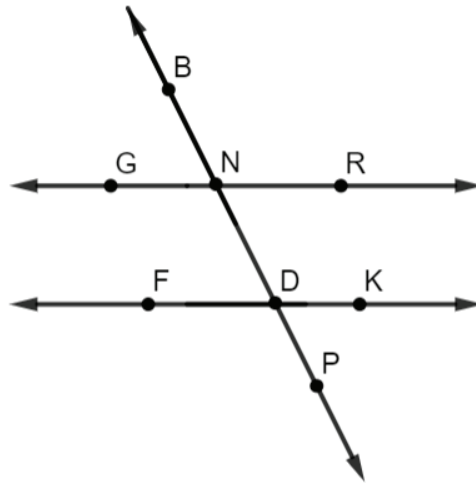


Statement	Reason

Proof B

Given: $\angle GND$ and $\angle PDK$ are supplementary.

Prove: $\overleftrightarrow{GR} \parallel \overleftrightarrow{FK}$

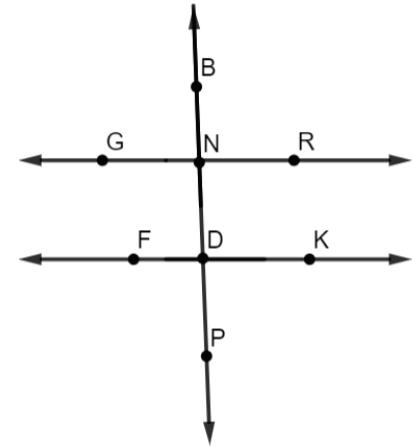


Statement	Reason

Proof C

Given: $\overleftrightarrow{BP} \perp \overleftrightarrow{GR}$, $\overleftrightarrow{GR} \parallel \overleftrightarrow{FK}$

Prove: $\overleftrightarrow{BP} \perp \overleftrightarrow{FK}$

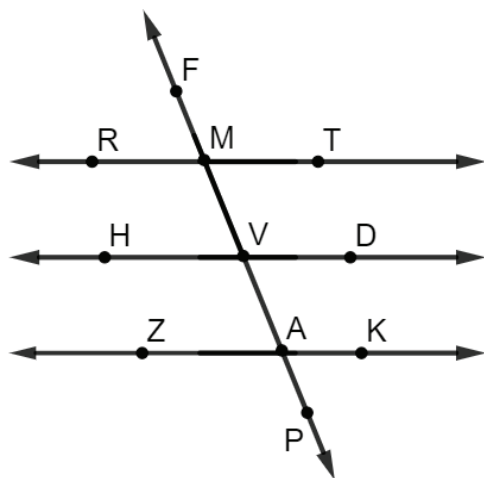


Statement	Reason

Proof D

Given: $\overleftrightarrow{RT} \parallel \overleftrightarrow{ZK}$,
 $\angle TMV \cong \angle AVD$

Prove: $\overleftrightarrow{HD} \parallel \overleftrightarrow{ZK}$

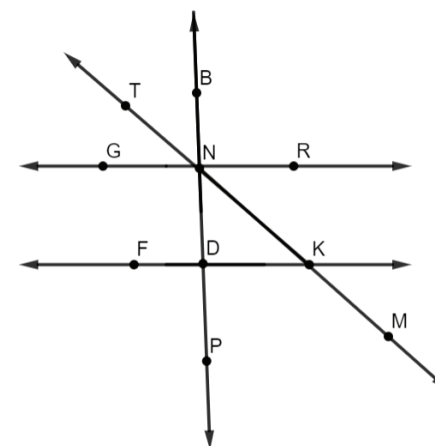


Statement	Reason

Proof E

Given: $\overleftrightarrow{BP} \perp \overleftrightarrow{GR}$ and
 $\overleftrightarrow{BP} \perp \overleftrightarrow{FK}$

Prove: $\angle DKM$ and
 $\angle GNT$ are
supplementary.

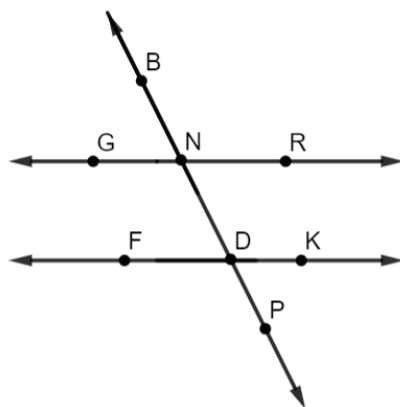


Statement	Reason

Proof F

Given: $\overleftrightarrow{GR} \parallel \overleftrightarrow{FK}$

Prove: $\angle GND$ and $\angle PDK$ are supplementary.

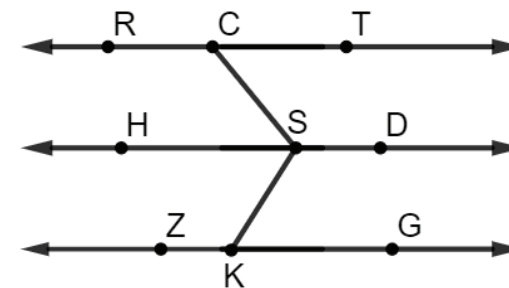


Statement	Reason

Proof G

Given: $\angle TCS \cong \angle HSC$,
 $\angle HSK \cong \angle SKG$

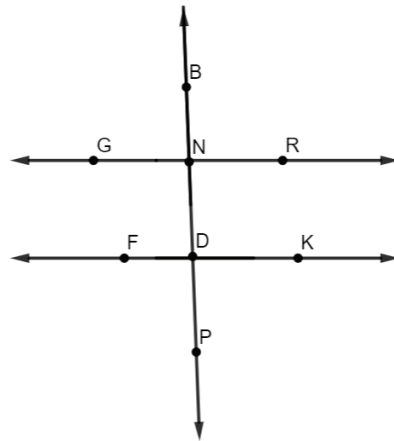
Prove: $\overleftrightarrow{RT} \parallel \overleftrightarrow{ZG}$



Statement	Reason

Proof H

Given: $\overleftrightarrow{GR} \perp \overleftrightarrow{BP}$ and
 $\overleftrightarrow{FK} \perp \overleftrightarrow{BP}$
 Prove: $\overleftrightarrow{GR} \parallel \overleftrightarrow{FK}$



Statement	Reason



3.9.3 Wrap-Up: Cool Down

1. Write an explanation to a classmate who might have missed today's lesson, in your own words, about what you've learned about angle relationship proofs. Include things that are easy to remember and common errors to watch out for.